Computer Viruses In UNIX networks

**Source: "Computer Viruses In Unix Networks"
Available at
www.cybersoft.com/whitepapers/papers/print/networks_print.html**

The Existence Of The Problem And Its Nature

The problem of software attacks exists in all operating systems. These attacks follow different forms according to the function of the attack. In general, all forms of attack contain a method of self preservation which may be propagation or migration and a payload. The most common method of self preservation in Unix is obscurity. If the program has an obscure name or storage location, then it may avoid detection until after its payload has had the opportunity to execute. Computer worms preserve themselves by migration while computer viruses use propagation. Trojan horses, logic bombs and time bombs protect themselves by obscurity.

While the hostile algorithms that have captured the general public's imagination are viruses and worms, the more common direct problem on Unix systems are Trojan horses and time bombs. A Trojan horse is a program that appears to be something it is not. An example of a Trojan horse is a program that appears to be a calculator or other useful utility which has a hidden payload of inserting a back door onto its host system. A simple Trojan horse can be created by modifying any source code with the addition of a payload. One of the most favourite payloads observed in the wild is "/bin/rm -rf / >/dev/null 2>&1" This payload will attempt to remove all accessible files on the system as a background process with all messages redirected to waste disposal. Since system security is lax at many sites, there are normally thousands of files with permission bit settings of octal 777. All files on the system with this permission setting will be removed by this attack. Additionally, all files owned by the user, their group or anyone else on the system whose files are write accessible to the user will be removed. This payload is not limited to use by Trojan horses but can be utilized by any